

2.5.17

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Question:

If

$$\mathbf{a} = \hat{i} - 7\hat{j} + 7\hat{k} \quad (0.1)$$

and

$$\mathbf{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}, \quad (0.2)$$

find a unit vector perpendicular to both the vectors $\mathbf{a}$ and $\mathbf{b}$.

Solution:

We want $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ such that

$$\mathbf{a}^T \mathbf{n} = 0, \quad (0.3)$$

$$\mathbf{b}^T \mathbf{n} = 0. \quad (0.4)$$

This gives the linear system

$$\begin{pmatrix} 1 & -7 & 7 \\ 3 & -2 & 2 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (0.5)$$

Step 1: Augmented matrix

$$\begin{pmatrix} 1 & -7 & 7 & 0 \\ 3 & -2 & 2 & 0 \end{pmatrix}. \quad (0.6)$$

Step 2: Row operations

$$R_2 \rightarrow R_2 - 3R_1 : \begin{pmatrix} 1 & -7 & 7 & 0 \\ 0 & 19 & -19 & 0 \end{pmatrix}, \quad (0.7)$$

$$R_2 \rightarrow \frac{1}{19}R_2 : \begin{pmatrix} 1 & -7 & 7 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad (0.8)$$

$$R_1 \rightarrow R_1 + 7R_2 : \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}. \quad (0.9)$$

Step 3: Solution

$$n_1 = 0, \quad (0.10)$$

$$n_2 - n_3 = 0 \Rightarrow n_2 = n_3. \quad (0.11)$$

Thus the general solution is

$$\mathbf{n} = t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}. \quad (0.12)$$

Step 4: Unit vector

Since

$$\left\| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\| = \sqrt{2}, \quad (0.13)$$

the unit vectors are

$$\hat{\mathbf{n}} = \pm \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad (0.14)$$

$$= \pm \frac{1}{\sqrt{2}} (\hat{j} + \hat{k}). \quad (0.15)$$

Vectors $\mathbf{a}$ (red), $\mathbf{b}$ (blue), and unit normal $\hat{\mathbf{n}}$ (green)

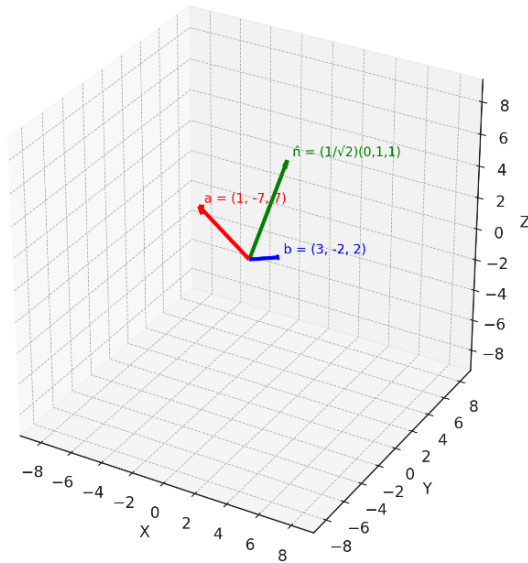


Fig. 0.1: Image Visual